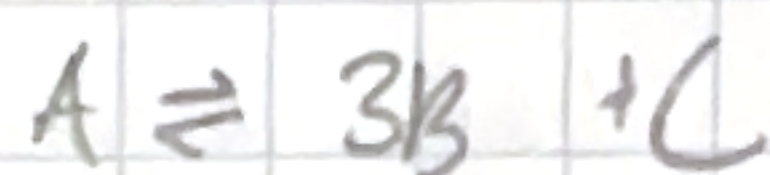
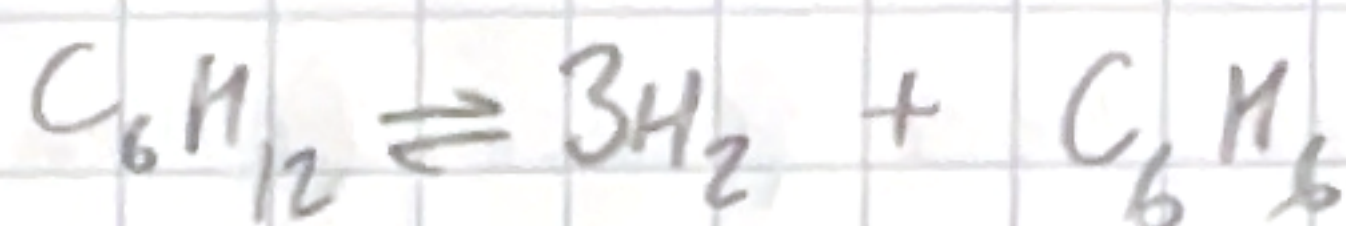


P6-1 c,d,e

P6-11

P6-1

3.



$$K_c = 0.001 \left(\frac{\text{mol}}{\text{dm}^3} \right)^3$$

$$\frac{dF_A}{dV} = r_A$$

$$\frac{dF_B}{dV} = 3r_B - R_B$$

$$R_B = k_c C_B$$

$$\frac{dF_C}{dV} = r_C$$

$$r_A = -k \left(C_A - \frac{C_B^3 C_C}{K_c} \right)$$

$$C_A = C_{T0} \frac{F_A}{F_T}$$

$$C_B = C_{T0} \frac{F_B}{F_T}$$

$$C_C = C_{T0} \frac{F_C}{F_T}$$

$$-r_A = k C_{T0} \left(\frac{F_A}{F_T} - \frac{C_{T0}}{K_c} \left(\frac{F_B}{F_T} \right)^3 \left(\frac{F_C}{F_T} \right) \right)$$

$$F_T = F_A + F_B + F_C$$

- more hydrogen is produced

4.

1) varying C_{B0} , v_0 , V_0

- C_{B0} - rate increases, conversion increases, concentration of C_B increases, concentration of C_A decreases, concentration of C_C increases

- v_0 - rate increase, conversion increases, concentration of C_A decreases, concentration of C_B increases, concentration of C_C decreases

- V_0 - conversion decreases, rate decreases, concentration of C_A increases, concentration of C_B decreases, concentration of C_D decreases

2) - increasing V_0 decreases concentration so reaction rate decreases

- higher V_0 increases C_A which increases reaction rate
- increasing C_{B0} increases rate because

$$r_A = -k C_A C_B$$

$$C_B = C_{T0} \left(\frac{F_B}{F_T} \right)$$

higher C_{B0} means higher C_B so r_A increases
 C_A slows down r_A when too much C_B

3) $A + B \rightleftharpoons C + D$

$$-r_A = k \left(C_A C_B - \frac{C_C C_D}{K_c} \right)$$

- increases reaction rate, conversion decreases

5.

1) α has the greatest effect on conversion

2) $p = 0.1$ Catalyst weight = 80 kg $C_{A0} = 0.046$
 $C_{B0} = 0.046$

$$p = \frac{P}{P_0} \Rightarrow 0.1 = \frac{P}{P_0} \quad P = 1 \text{ atm}$$

$$\frac{dp}{dW} = \frac{-\alpha}{2\gamma} \left(\frac{F_T}{F_{T0}} \right) \left(\frac{T}{T_0} \right)$$

$$C_A = \frac{F_A}{V_0}$$

$$C_A = C_{T0} \left(\frac{F_A}{F_T} \right) p$$

$p = 0.1$ at high α - not included in graph

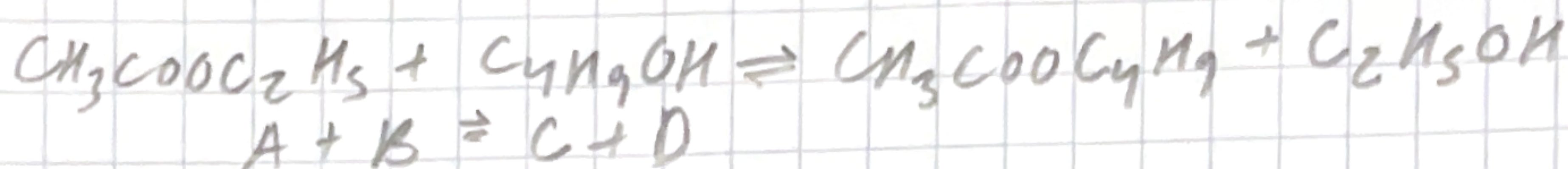
- higher α
- larger catalyst weight

3) - α reduces pressure so large drop in pressure

- higher catalyst weight decreases pressure

P6-11

semibatch reactor



$$T = 300\text{K}$$

$$K_c = 1.08$$

$$-r_A = 9 \cdot 10^{-5} \frac{\text{dm}^3}{\text{mol} \cdot \text{s}}$$

$$V_0 = 200 \frac{\text{dm}^3}{\text{s}}$$

$$v_0 = 0.05 \frac{\text{dm}^3}{\text{s}}$$

$$C_{B,F} = 10.93 \frac{\text{mol}}{\text{dm}^3}$$

$$C_{A0} = 7.72 \frac{\text{mol}}{\text{dm}^3}$$

3. $C_D \approx 0$

$$-r_A = k (C_A C_B)$$

$$-r_A = k \left(C_A C_B - \frac{C_C C_D}{K} \right)$$

4. Varying F_{B0} $\Rightarrow$ conversion of A increases

N_{A0} - decreases conversion

v_0 - conversion decreases, rate decreases

6. How would results change if reaction was reversible?

- requires understanding reversible vs irreversible rates.